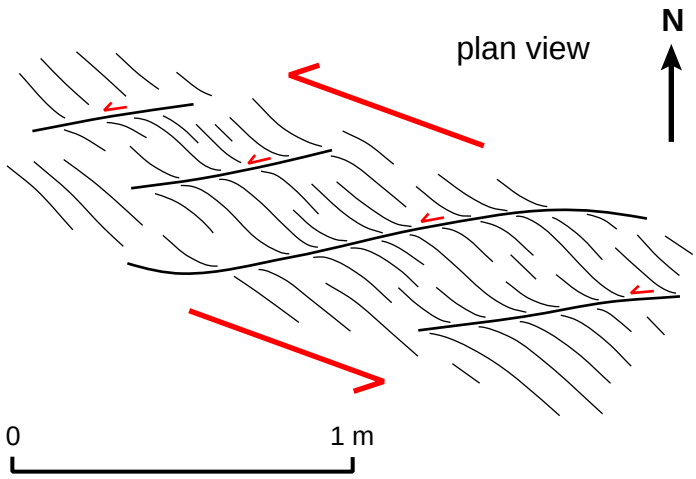
SHEAR ZONE EXERCISE

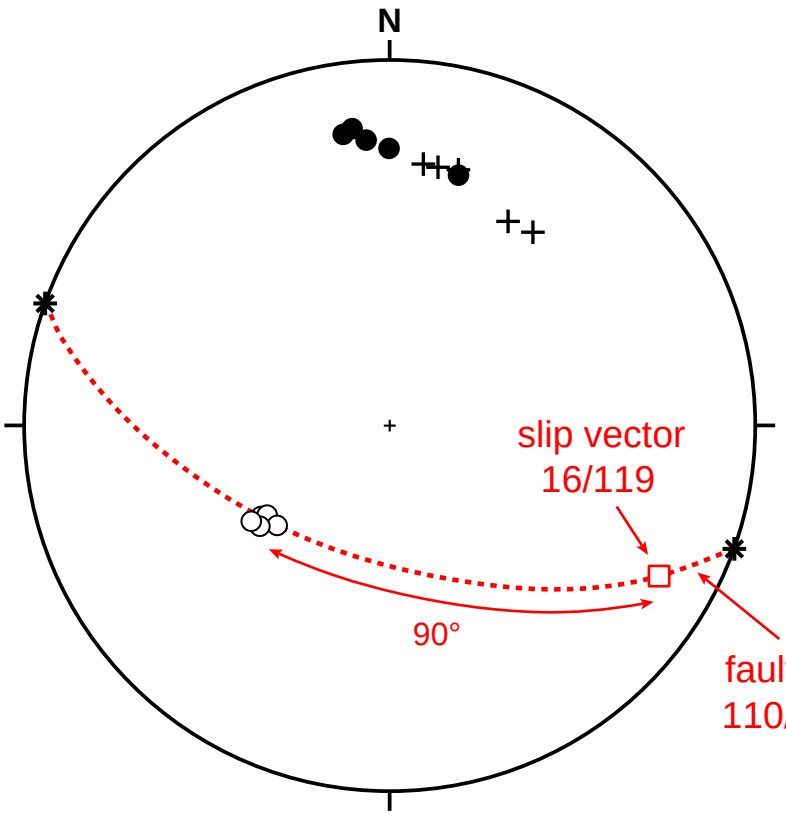
In an area of low topography there is a 20 m wide fault zone filled with a clay-bearing fault gouge. The fault zone trends 110° but dips at unknown angle. The clay-bearing fault gouge in the centre of the fault zone has two fabrics: a penetrative cleavage and a spaced set of shear bands. The angular relationship between these two fabrics is illustrated in the field sketch below

The following measurements were obtained at different places within the fault zone:

locality	cleavage	shear band
1	097/62S	081/70S
2	120/55S	090/66S
3	126/56S	105/61S
4	100/62S	085/68S
5	105/62S	083/72S



Determine the orientation of the fault plane, fault slip vector, and the sense of movement on the fault.



For the record,
dominant P-foliation orientation: 126/56S
dominant Riedel orientation: 081/70S

+ poles to cleavage = P-foliation
● poles to shear bands = Riedel shears
✱ strike of fault zone
○ cleavage / shear band intersections

Fault movement is sinistral and normal